Federal Urdu University
of Arts, Science and Technology

SmartDine

Food Recommendation Agent

Course: AI

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1. Introduction.

SmartDine is a **Food Recommendation Agent** , an intelligent software agent designed to recommend suitable food items to customers based on their preferences, such as cuisine, budget, dietary preference, spice level, and previous choices.

2.Problem Formulation.

Textual Representation.

Environment.

The environment includes the restaurant, menu, food items, prices, categories, cuisines, and food availability.
The agent uses this information to make suitable food recommendations.

Sensors (Input).

Sensors collect customer preferences, such as budget, cuisine, food category, spice level, and dietary requirements.
They also receive relevant information from the restaurant menu.

Processing

The agent analyzes the customer's preferences and compares them with available menu items. It evaluates the matching options and selects the most suitable food item.

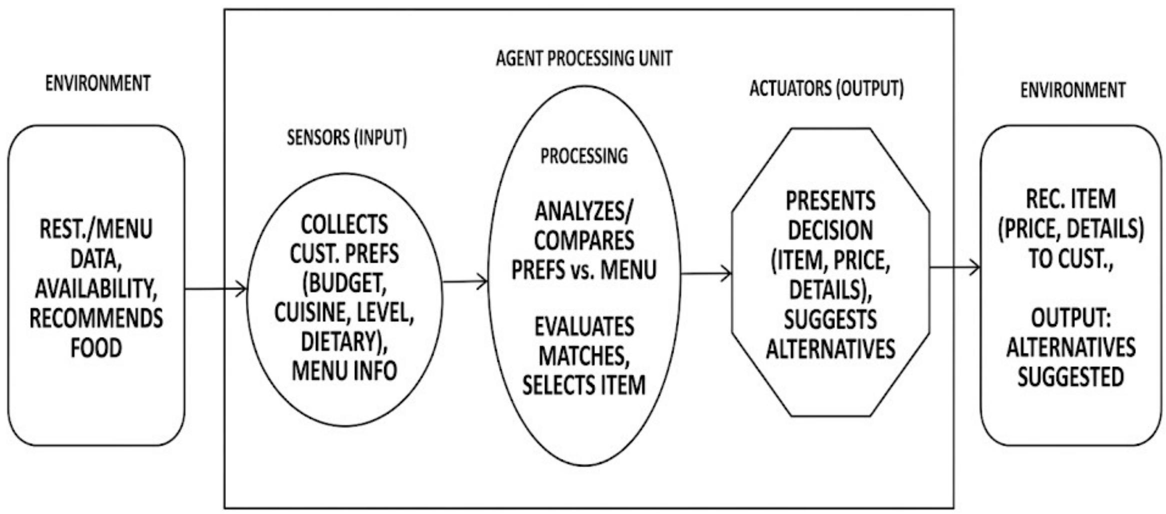
Actuators (Output)

Actuators present the agent's decision by displaying the recommended food item, price, and relevant details.
The agent can also suggest suitable alternatives.

Example. A customer wants Italian food under Rs. 800 with medium spice. The sensors collect these preferences and the processing unit compares them with the restaurant's menu. After evaluating the available options, the agent selects Chicken Pasta for Rs. 750 as the best match. The actuator then displays the recommendation, price, and food details to the customer.

Flow:
Customer Preferences → Sensors → Processing → Actuator → Chicken Pasta Recommendation

Graphical Representation.



3.PEAS Framework.

Component	Evaluation
P – Performance Measure	Accuracy of recommendation, customer preference matching, budget matching, food availability, and customer satisfaction.
E – Environment	Restaurant, food menu, food items, prices, categories, cuisines, and food availability.
A – Actuators	Display recommended food, show price and details, and suggest alternative food items.
S – Sensors	Customer budget, cuisine preference, food category, spice level, dietary requirements, and menu information.

4.Environmental Properties

Property	Type	Reason
Observability	**Partially Observable**	The agent knows the customer's given preferences but cannot know their complete taste or mood.
Determinism	**Deterministic**	For the same customer preferences and menu data, the agent follows the same recommendation rules.
Episodic /	**Episodic**	Each recommendation request is treated as a separate

Property	Type	Reason
Sequential		episode.
Static / Dynamic	**Dynamic**	Menu items, prices, and food availability can change.
Discrete / Continuous	**Discrete**	Inputs such as cuisine, category, spice level, and dietary preference have defined values.
Single / multi-Agent	**Single-Agent**	Only one intelligent agent, the Food Recommendation Agent, makes the recommendation.
Known / Unknown	**Partially Known**	The menu information is known, but the customer's complete preferences may be unknown.